Supplemental Instruction Lesson Plan

Session Date: 10 FEB 2022
Course: CSCI160 (Computer Science I)

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Lesson: Using Python operations

Background: Modern technology people use everyday rely on computers and our ability to use computers to break down data and providing new useful information. Vending machines and cash registers can quickly calculate the most efficient way to return change. Today, we are going to familiarize ourselves on using Python operations to solve some of these practical problems.

Practical Applications:

- Create/identify patterns
- Repeating tasks and intervals
- Return correct change at cash register
- Path-finding AI

Agenda:

1. Python operations
2. Group Activities

1. Python Operations

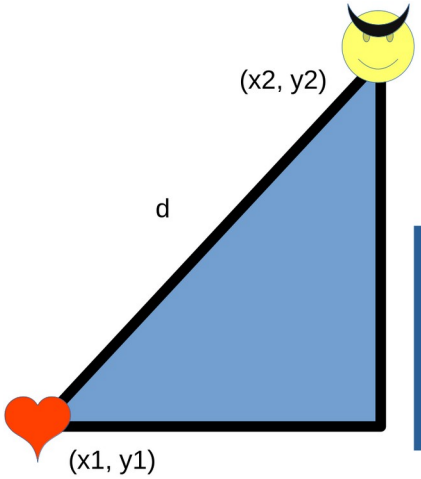
I. Exponentiation

II. Floor Division

III. Modulus

2. Group Activities

I. You're making a video game in Python. You have a player and an AI. The AI has an attack that he will use when it is within a certain distance from the player. You already implemented the ability to get the AI's range and each (x, y) coordinates (using the input methods below). Print the actual distance between the players so you can see if the AI can attack.



$$d=\sqrt{(x2-x1)^2+(y2-y1)^2}$$

```
AI_range = input('')
x1 = input('')
y1 = input('')
x2 = input('')
y2 = input('')

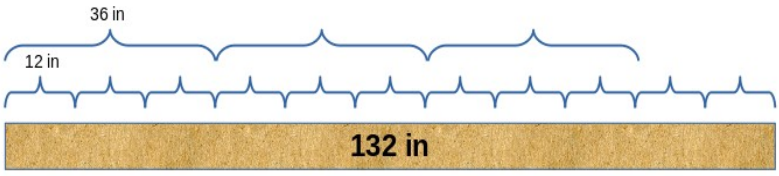
# code
```

Figure 1: Exponentiation demo graphic with distance formula and starter code

Example Input:
2
0
0
3
4

Output:
Enemy range is 2
Distance is 5.0

II. Your friend is ordering a comically long piece of wood for your yardstick factory, AKA your garage. The piece of wood is already the correct dimensions for a yardstick or rulers, so all you need to do is cut the wood lengthwise. Use python to get the maximum number of rulers you can make and the maximum number of yardsticks you can make if the piece of wood is x inches long (get x from input).



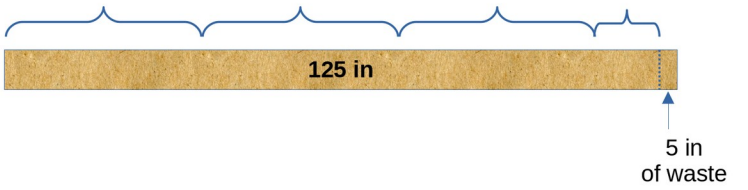

```
1  inches = input('')
2  inches = int(inches)
3
4  # code
5
6  print('Possible yardsticks:', yardsticks)
7  print('Possible rulers:', rulers)
```

Figure 2: Floor Division demo graphic with starter code. This board can make either a maximum of 3 yardsticks or a maximum of 11 rulers.

Example Input:
132

Output:
Possible yardsticks: 3
Possible rulers: 11

III. Now you want to make as many yardsticks as you can, but you will make as many rulers as you can with the leftover wood.




```
8
9   # code
10
11  print('Max yardsticks:', max_yardsticks)
12  print('Leftover rulers:', leftover_rulers)
13  print('Leftover waste:',leftover_waste, 'inches')
14
```

Figure 3: Modulus demo graphic. This board can make a maximum of 3 yardsticks, with 1 rulers and 5 inches of waste leftover.

Example Input:
125

Output:
Max yardsticks: 3
Leftover rulers: 1
Leftover waste: 5 inches

Resources:

- https://www.w3schools.com/python/python_operators.asp